

Generative Modeling with Optimal Transport Maps

Paper review: Rout, Korotin & Burnaev, ICLR 2022

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Key Formulas: AE-OT vs OT-map (ICLR 2022)

This summary compares the core mathematical formulations of AE-OT (Autoencoder + Optimal Transport) and the OT-map method from “Generative Modeling with Optimal Transport Maps” (ICLR 2022).

AE-OT: Autoencoder + Optimal Transport

Phase 1 – Autoencoder training (supervised reconstruction):

$$\min_{E,D} \mathbb{E}_{y \sim \nu} [\|D(E(y)) - y\|^2]$$

- $y \in \mathbb{R}^D$: real image
- $z = E(y) \in \mathbb{R}^H$: latent representation

Phase 2 – Discrete OT between noise and latent codes:

Let $x_i \sim \mu$ (noise), and $z_j = E(y_j) \sim E_{\#}\nu$.

Option 1: Monge OT (with permutation):

$$\min_T \sum_{i=1}^n \|T(x_i) - z_{\pi(i)}\|^2$$

Option 2: Entropic Kantorovich OT with barycentric projection:

$$\min_P \sum_{i,j} P_{ij} \|x_i - z_j\|^2 + \varepsilon \sum_{i,j} P_{ij} \log P_{ij}$$

$$T(x_i) = \sum_j P_{ij} z_j \quad (\text{barycenter})$$

Final generation:

$$\hat{y} = D(T(x))$$

OT-map (ICLR 2022)

Setting: Learn transport map $G(x) = T(Q(x))$ directly in the image space $\mathbb{R}^D$, without any encoder.

Min–max formulation (dual of OT):

$$\min_{\psi} \max_G [\mathbb{E}_{x \sim \mu} (\langle Q(x), G(x) \rangle - \psi(G(x))) + \mathbb{E}_{y \sim \nu} \psi(y)]$$

- $Q(x) \in \mathbb{R}^D$: embedded noise (e.g., via upsampling)
- $G(x) \in \mathbb{R}^D$: image generated from noise
- $\psi(y)$: Kantorovich potential (convex function)

Optimal transport geometry:

$$G(x) \approx \arg \max_y (\langle Q(x), y \rangle - \psi(y)) \quad \Rightarrow \quad G(x) \in \partial \psi^*(Q(x))$$

No decoder needed — generation is direct:

$$\hat{y} = G(x)$$

Comparison

Feature	AE-OT	OT-map (this paper)
Space of transport	Latent space $\mathbb{R}^H$	Ambient space $\mathbb{R}^D$
Needs autoencoder?	Yes	No
Uses encoder/decoder?	Yes: E, D	No
Transport computed via	Discrete OT (Monge or Sinkhorn)	Min–max dual OT formulation
Decoding step	$\hat{y} = D(T(x))$	$\hat{y} = G(x)$ (direct output)
Type of supervision	Point set alignment	Distribution-level alignment

Derivation of the OT-map Objective

1. Monge’s Problem (Primal Formulation)

Let μ and ν be two probability measures defined on metric spaces $\mathcal{X}, \mathcal{Y} \subset \mathbb{R}^d$. The original problem posed by Monge (1781) is:

$$\inf_{T_{\#}\mu=\nu} \int_{\mathcal{X}} c(x, T(x)) d\mu(x)$$

where $T : \mathcal{X} \rightarrow \mathcal{Y}$ is a measurable map that pushes forward μ to ν (i.e., $T_{\#}\mu = \nu$), and $c(x, y)$ is a ground cost function.

Remark: Monge’s problem may be ill-posed; it does not allow mass splitting and may lack a solution if, e.g., μ is a Dirac delta and ν is not.

2. Kantorovich’s Relaxation

Kantorovich (1942) relaxed the problem by allowing probabilistic couplings π between μ and ν . The new problem becomes:

$$\inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y)$$

where $\Pi(\mu, \nu)$ is the set of all joint distributions with marginals μ and ν .

3. Dual Formulation and c -Transform

The Kantorovich dual problem is:

$$\sup_{u,v} \left\{ \int_{\mathcal{X}} u(x) d\mu(x) + \int_{\mathcal{Y}} v(y) d\nu(y) : u(x) + v(y) \leq c(x, y) \right\}$$

This is a linear program over functions u, v . To work more effectively with this constraint, we define the **c -transform**:

Definition: c -Transform

Given $v : \mathcal{Y} \rightarrow \mathbb{R}$, the c -transform of v is:

$$v^c(x) := \inf_{y \in \mathcal{Y}} \{c(x, y) - v(y)\}$$

Analogously, the c -transform of u is:

$$u^c(y) := \inf_{x \in \mathcal{X}} \{c(x, y) - u(x)\}$$

Using the c -transform, we can rewrite the dual in terms of a single potential:

$$\sup_v \left\{ \int_{\mathcal{X}} v^c(x) d\mu(x) + \int_{\mathcal{Y}} v(y) d\nu(y) \right\}$$

This is equivalent to:

$$\sup_u \left\{ \int_{\mathcal{X}} u(x) d\mu(x) + \int_{\mathcal{Y}} u^c(y) d\nu(y) \right\}$$

4. Specialization to Quadratic Cost and Convex Potentials

Let us now consider the quadratic cost (the OT value is then $\frac{1}{2}W_2^2$):

$$c(x, y) = \frac{1}{2}\|x - y\|^2$$

Why convex functions? Brenier's theorem (1991) states: if μ is absolutely continuous and $c(x, y) = \frac{1}{2}\|x - y\|^2$, then the optimal transport map T^* is the gradient of a convex function:

$$T^*(x) = \nabla \varphi(x)$$

This justifies parameterizing the OT map as the gradient of a convex potential.

5. Reformulating the Dual using Convex Potentials

Let us define:

$$\psi(y) := \frac{1}{2}\|y\|^2 - v(y)$$

Then we obtain:

$$v(y) = \frac{1}{2}\|y\|^2 - \psi(y)$$

and so:

$$\begin{aligned}
v^c(x) &= \inf_y \left\{ \frac{1}{2} \|x - y\|^2 - v(y) \right\} \\
&= \inf_y \left\{ \frac{1}{2} \|x - y\|^2 - \left(\frac{1}{2} \|y\|^2 - \psi(y) \right) \right\} \\
&= \inf_y \left\{ \frac{1}{2} \|x\|^2 - \langle x, y \rangle + \psi(y) \right\} \\
&= \frac{1}{2} \|x\|^2 - \sup_y \{ \langle x, y \rangle - \psi(y) \} \\
&= \frac{1}{2} \|x\|^2 - \psi^*(x)
\end{aligned}$$

Thus, the dual becomes:

$$\frac{1}{2} W_2^2(\mu, \nu) = \int \frac{1}{2} \|x\|^2 d\mu(x) + \int \frac{1}{2} \|y\|^2 d\nu(y) - \inf_{\psi \text{ convex}} \left\{ \int \psi^*(x) d\mu(x) + \int \psi(y) d\nu(y) \right\}$$

6. Final Form: Saddle Point Optimization

Using the Fenchel–Legendre duality:

$$\psi^*(x) = \sup_{y \in \mathbb{R}^d} \{ \langle x, y \rangle - \psi(y) \}$$

So:

$$\int \psi^*(x) d\mu(x) = \sup_T \int \langle x, T(x) \rangle - \psi(T(x)) d\mu(x)$$

The final expression is:

$$\frac{1}{2} W_2^2(\mu, \nu) = \int \frac{1}{2} \|x\|^2 d\mu(x) + \int \frac{1}{2} \|y\|^2 d\nu(y) - \inf_{\psi \text{ convex}} \sup_T \left\{ \int \langle x, T(x) \rangle - \psi(T(x)) d\mu(x) + \int \psi(y) d\nu(y) \right\}$$

7. Optimization Objective for Generative Modeling

Letting the constant terms be absorbed, we minimize the functional:

$$\min_{\psi} \max_T \left\{ \int \langle x, T(x) \rangle - \psi(T(x)) d\mu(x) + \int \psi(y) d\nu(y) \right\}$$

This is the saddle point problem solved in the ICLR 2022 paper, where both T and ψ are parameterized by neural networks.